

Satellite Navigation

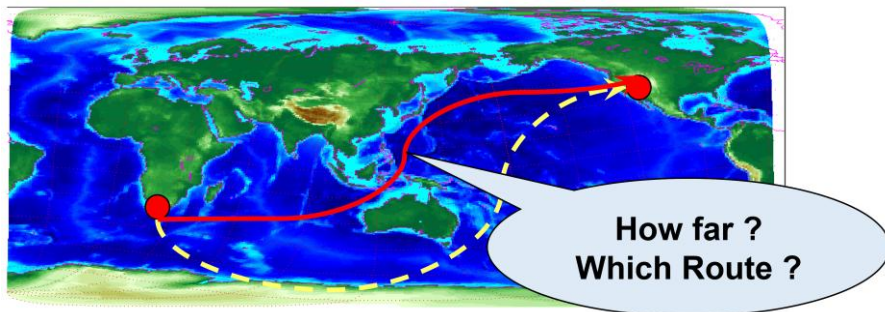
Lecture 9

Dr Qiao Cheng
qiao.cheng@lancaster.ac.uk

CNSCC.160:
Fundamentals of Communication Systems

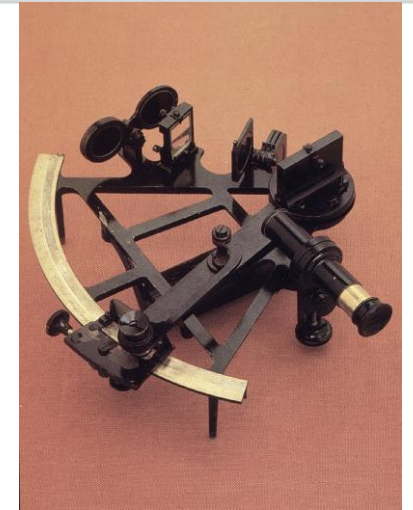
Problem

- How can I know my location precisely ?
 - In any weather condition
 - At any time
 - Everywhere on earth (at least outdoors!)
- How to navigate to the destination? ?
 - Guidance or Navigation



Development of Satellite Navigation Systems

- **Landmark-based Navigation**
 - Marking of trails
 - Using **Stones, Trees, Monuments** etc
 - **Limited** - Local use
- **Celestial-based Navigation**
 - **Sun, Moon** and **Stars**
 - **Complicated**, Works only at Clear Night
- **Sensors-based Navigation**
 - **Magnetic compass, sextant, Gyroscope, Accelerometer**
 - **Complicated**, Errors accumulate quickly



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- Ground-based radio navigation techniques
 - LORAN, OMEGA
 - Subject to Radio Interference, Jamming, Limited Coverage
 - Satellite-based Navigation or GNSS
 - TRANSIT, GPS, GLONASS, GALILEO, QZSS, BEIDOU (COMPASS), IRNSS
 - Global, Difficult to Interfere or Jam, High Accuracy & Reliability

Global Navigation Satellite System

- Global Navigation Satellite System (GNSS) is the standard generic term for all navigation satellites systems like GPS, GLONASS, GALILEO, BeiDou, QZSS, NAVIC.
- **Global Constellation**
 - GPS USA
 - GLONASS, Russia
 - Galileo, Europe
 - BeiDou(COMPASS), China
- **Regional Constellation**
 - QZSS, Japan
 - NAVIC (IRNSS), India

Key Global Systems - GPS and GLONASS Overview

- **GPS**
 - Developed by the U.S
 - Fully operational since 1993
 - Consists of at least 24 satellites in various orbital planes to ensure global coverage
- **GLONASS**
 - Developed by the Soviet Union, started in 1982
 - **Restoration:** Fully restored and modernized by Russia in the 2000s
 - Comprises 21 active satellites in three orbital planes, optimized for high latitude performance

Applications of Satellite Navigation - General Introduction

- **Everyday Life**
 - Used in [personal navigation devices](#), smartphones, and in-car navigation systems to provide real-time positioning and directions
- **Commercial Applications**
 - Critical in logistics for [tracking shipments and optimizing routes](#) to increase operational efficiencies
- **Scientific Research**
 - Used in [environmental monitoring](#), weather forecasting, and space science
- **Safety and Emergency Services**
 - Vital for search and rescue operations, [disaster response coordination](#), and [law enforcement tracking](#)
- **Future Prospects**
 - Growing applications in [autonomous driving technologies](#), advanced robotics, and potential [expansions in unmanned aerial vehicle usage](#)

Historical Development of GPS - Timeline and Key Milestones

- **1973**: Conception by the U.S
- **1978**: Launch of the first GPS satellite, NAVSTAR
- **1983**: After the **Soviet Union** shot down a Korean airliner that strayed into its airspace, President Reagan opened GPS to civilian use to prevent such tragedies
- **1993**: Achieves full operational capability with a constellation of 24 satellites
- **2000**: Selective Availability, **which intentionally degraded civilian GPS accuracy**, was **discontinued** under President Bill Clinton, significantly improving the utility of GPS for civilian applications
- **2005-2010**: Introduction of **modernized satellites with new capabilities**, including a second civilian signal to enhance user performance and increase resistance to interference

Historical Development of GLONASS - Timeline and Key Milestones

- **1976:** Development initiated by the Soviet Union as a response to the GPS system
- **1982:** The first GLONASS satellite was launched
- **1995:** A full constellation of 21 satellites was achieved, providing global coverage
- **2001:** Following a decline in the system's capabilities due to post-Soviet economic issues, the Russian government re-committed to fully restoring GLONASS
- **2011:** GLONASS regained full operational capability with the launch of additional satellites
- **2017-Present:** Continual upgrades with GLONASS-K satellites that offer better accuracy, extended operational lifespan, and additional civilian frequency bands

Timing Signals and Their Importance - How Timing Affects Accuracy

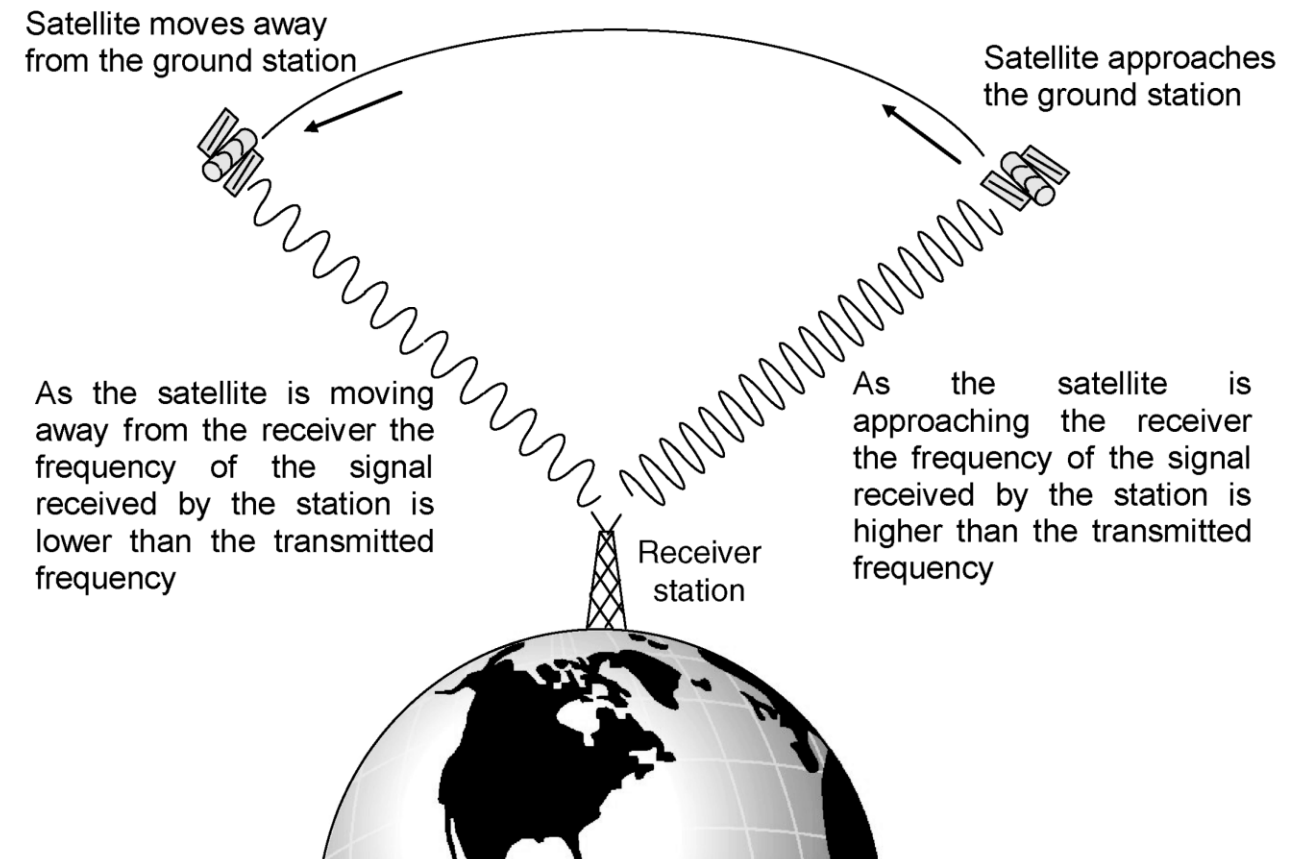
- **Role of Timing**
 - Critical for calculating distances; even nanosecond errors can lead to significant positional inaccuracies
- **Atomic Clocks in Satellites**
 - These clocks provide the time stamp for the signals sent from satellites
- **Receiver Clocks**
 - Generally less accurate; receivers use data from multiple (4th) satellites to correct their own clocks
- **Impact on Positioning Accuracy**
 - The correct timing leads to accurate distance measurements, which are crucial for precise positioning
- **Example**
 - A timing error of 1 microsecond could result in a positioning error of up to 300 meters, underscoring the need for precision

Signal Propagation - Paths from Satellite to Receiver

- **Path Details**
 - Signals travel through the vacuum of space and the Earth's atmosphere before reaching the receiver
- **Influences on Signal Speed and Trajectory**
 - **Ionosphere and Troposphere Effects:** The atmospheric conditions can delay and refract the signals
 - **Multipath Errors:** The signals can bounce off buildings or other large objects, causing errors in signal travel time and thus in position calculations
- **Mitigating Signal Propagation Issues**
 - Use of **advanced receiver technologies like multi-frequency receivers and techniques** to filter out noise and correct errors

Doppler Effect based Satellite Navigation Systems

- USA
 - Transit
 - 1960-1996
- USSR
 - Kosmos
 - 1967-1995



Trilateration-based Satellite Navigation Systems

The Global Positioning System (GPS) is a **Trilateration-based Satellite Navigation Systems**, which:

- was developed as a military system by the U.S. DoD in **the 1970s**
- was initiated in **1978** (first satellite launch)
- is fully **operational since 1995** (24 satellites, now >30)

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- is available for civilian use since 1996 (**selective availability until 2000**)
 - works ‘**anywhere**’ in the world under any weather conditions
 - serves an **unlimited number of users**
 - requires **no subscription fees** or set-up charges
 - consists of **three segments**
- 

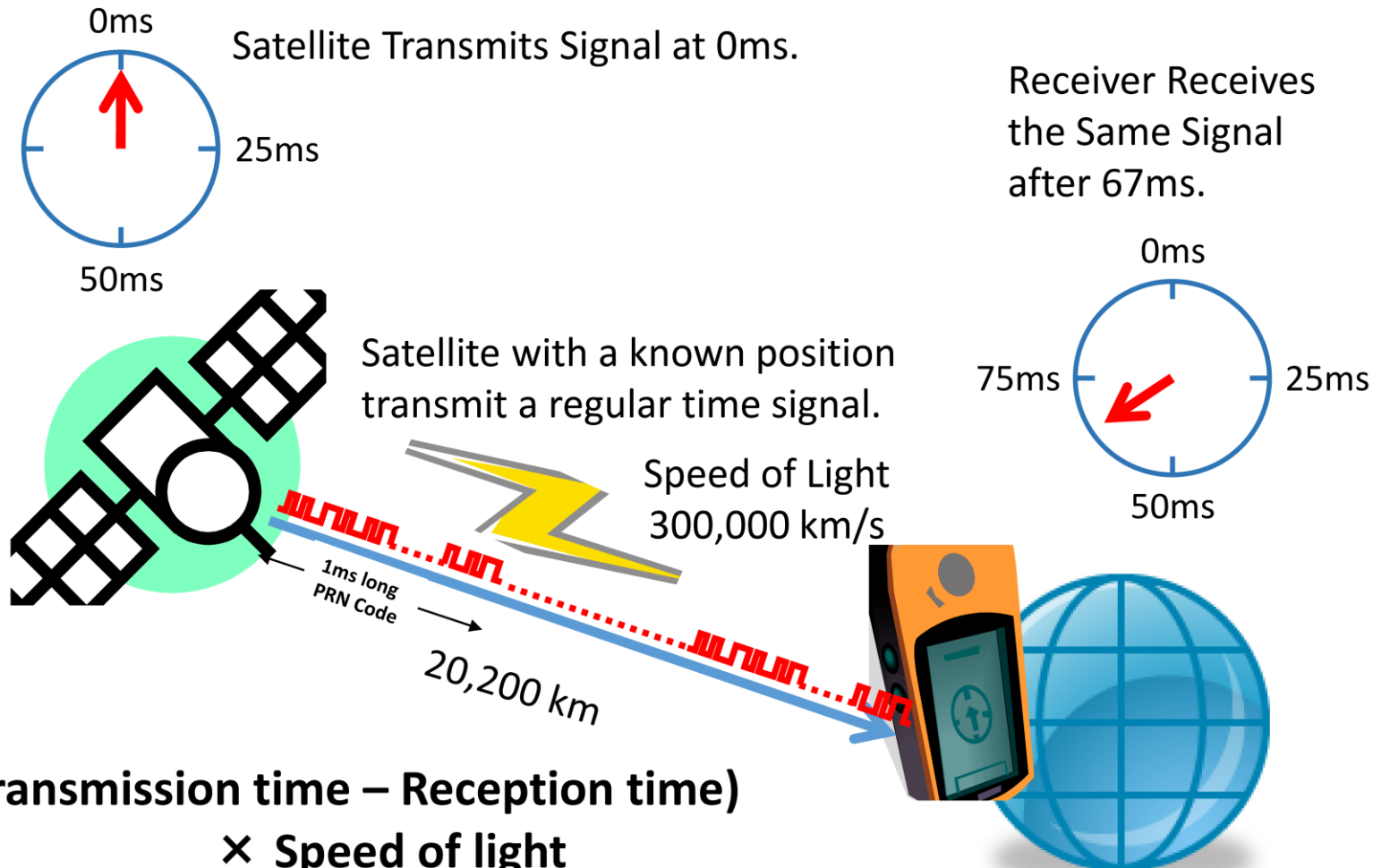
GPS - How does it work?

- GPS satellites circle the earth twice a day in a very precise orbit and transmit signal information to Earth.
- GPS receivers take this information and use **trilateration** (三边测量) to calculate the user's exact location.

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- Essentially, the GPS receiver compares the **time a signal was transmitted** by a satellite with the **time it was received**.
- The time difference tells the GPS receiver how far away the satellite is.
- With distance measurements from several satellites, the receiver can determine the user's position.

Determine the Distance



GPS Signal Power

- GPS satellites are about 22,000km away
- Transmit power is about 30W
 - This power reduces by $1/\text{distance}^2$
 - Power when received at the receiver is reduced by 10^{16} times.
 - This is similar to seeing a 30W bulb 22,000Km far
- GPS signals in the receiver is about 10^{-16} Watt, which is below the thermal noise

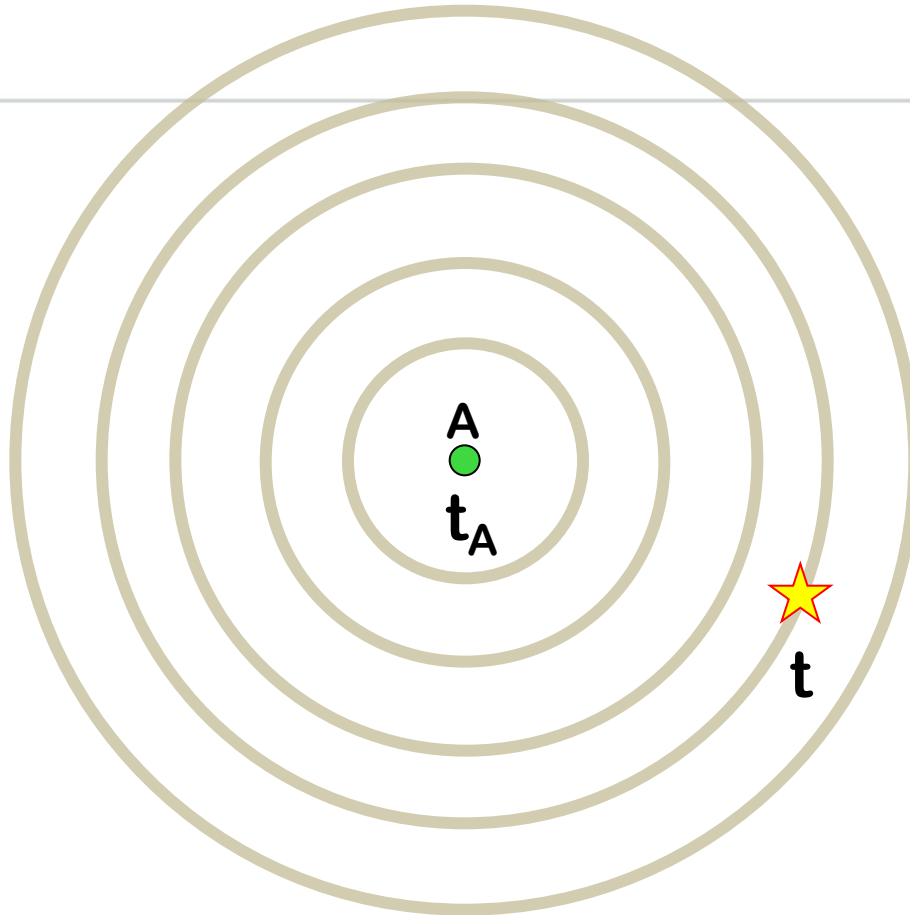
2D user position



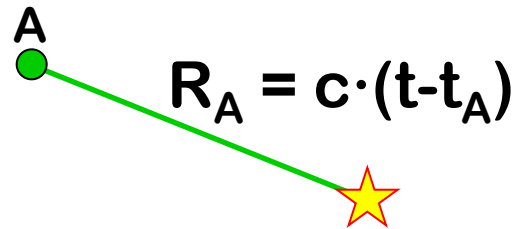
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A small green dot representing a point in a 2D space.

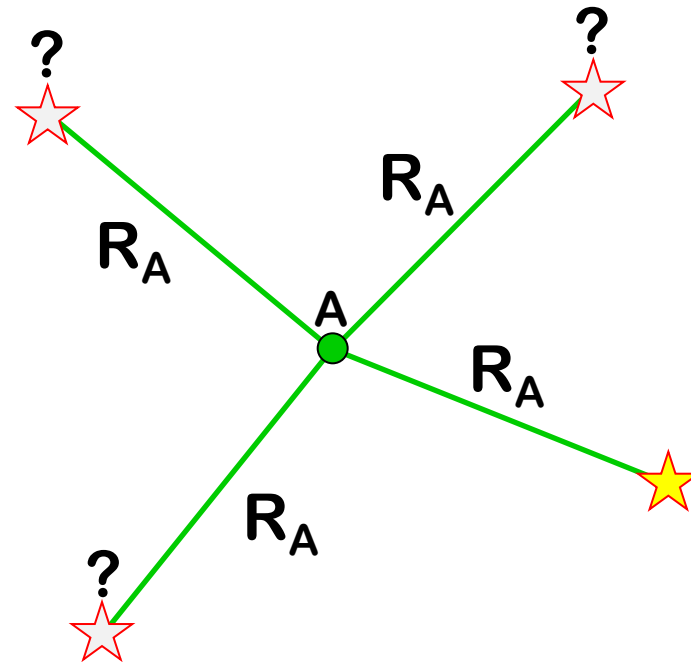
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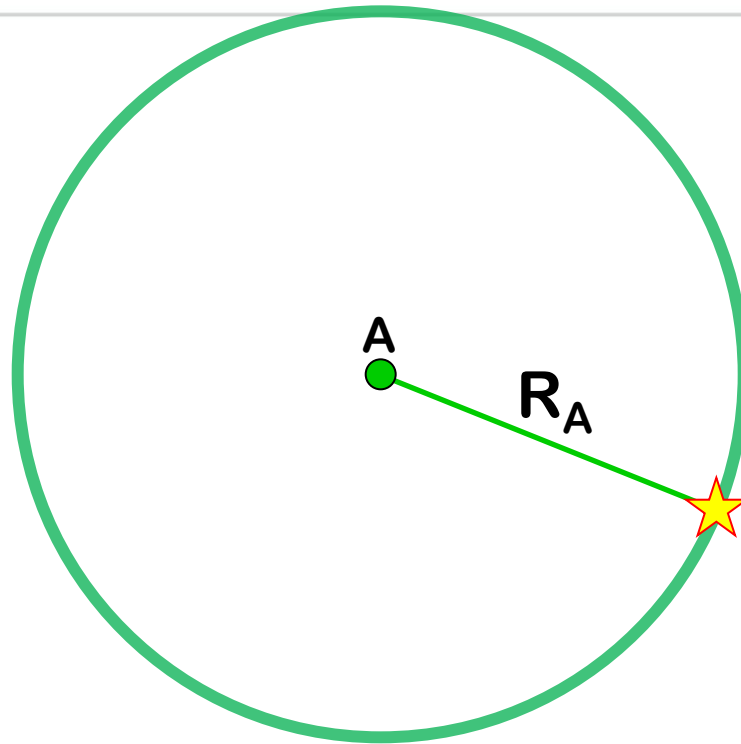
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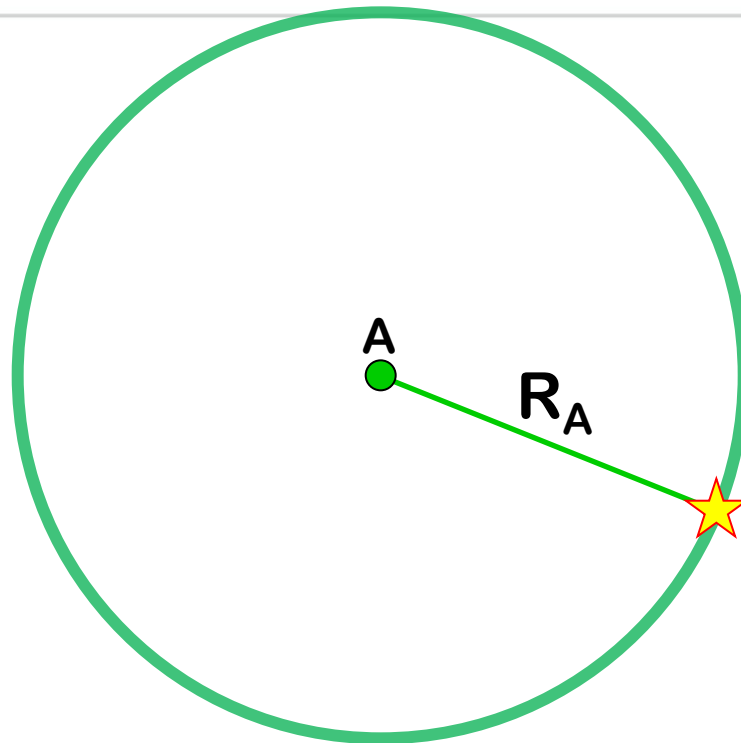
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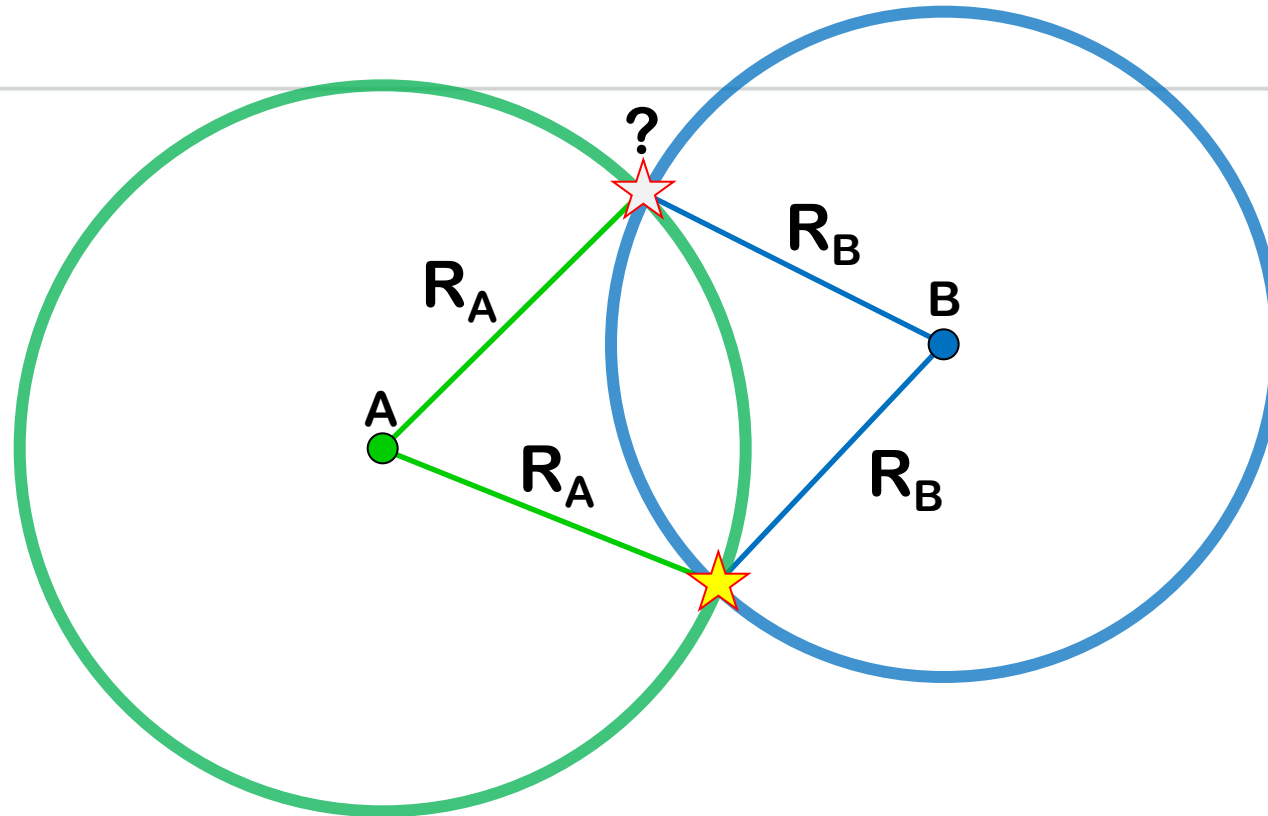
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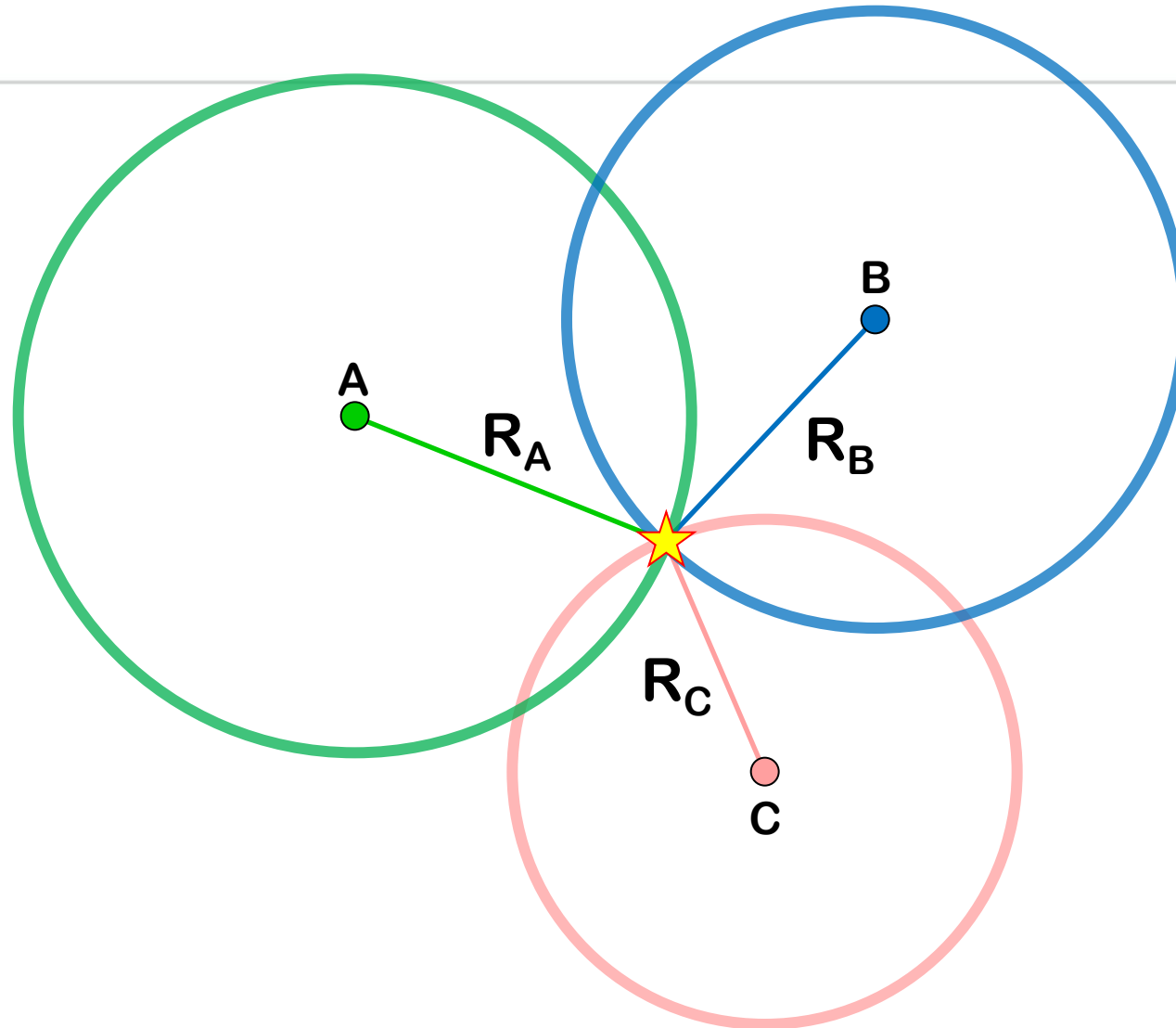
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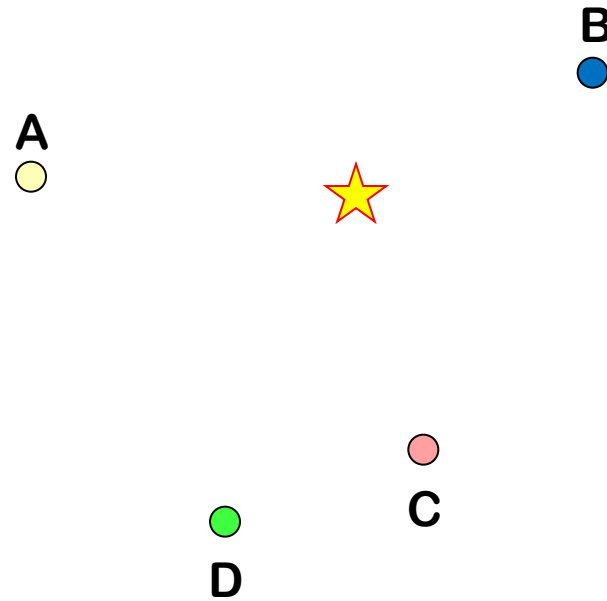
2D user position



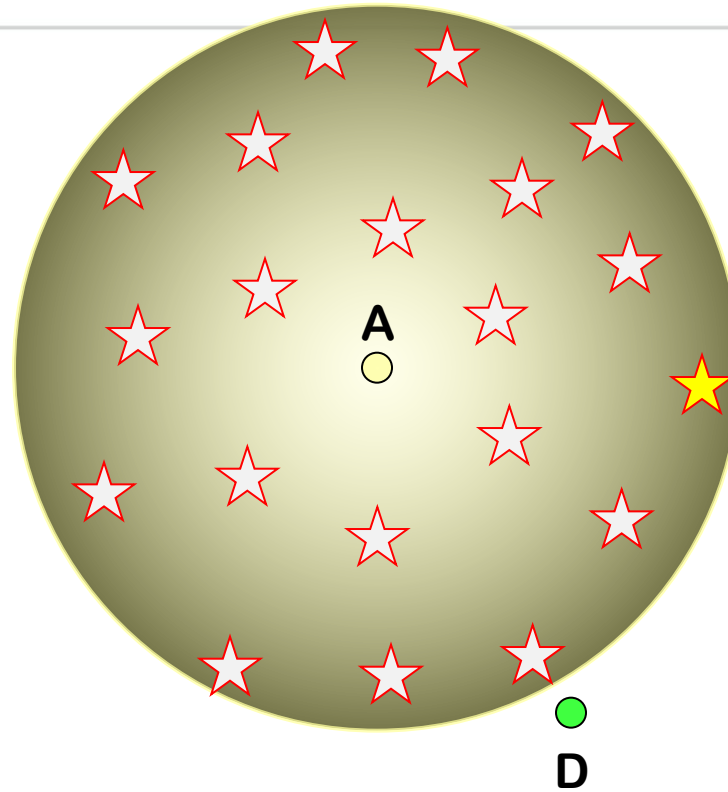
2D user position



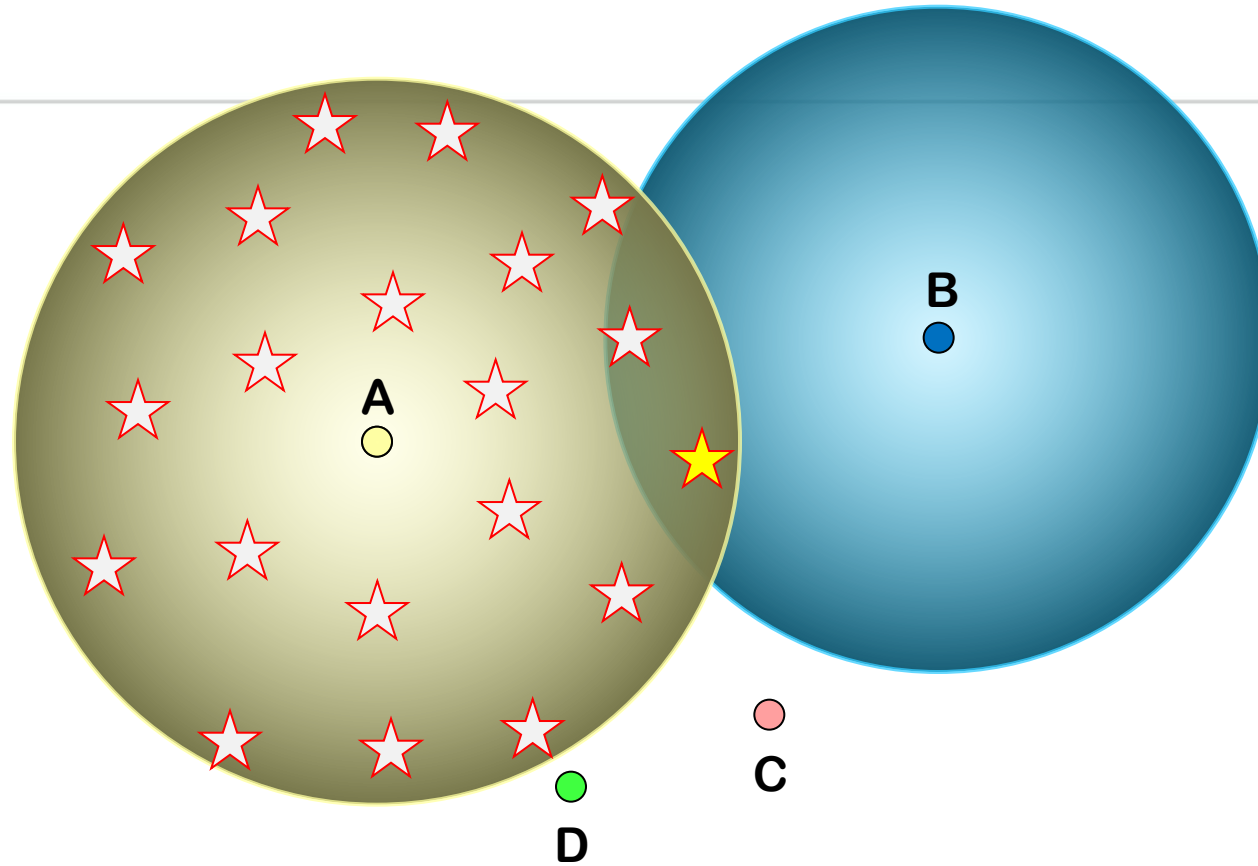
3D user position



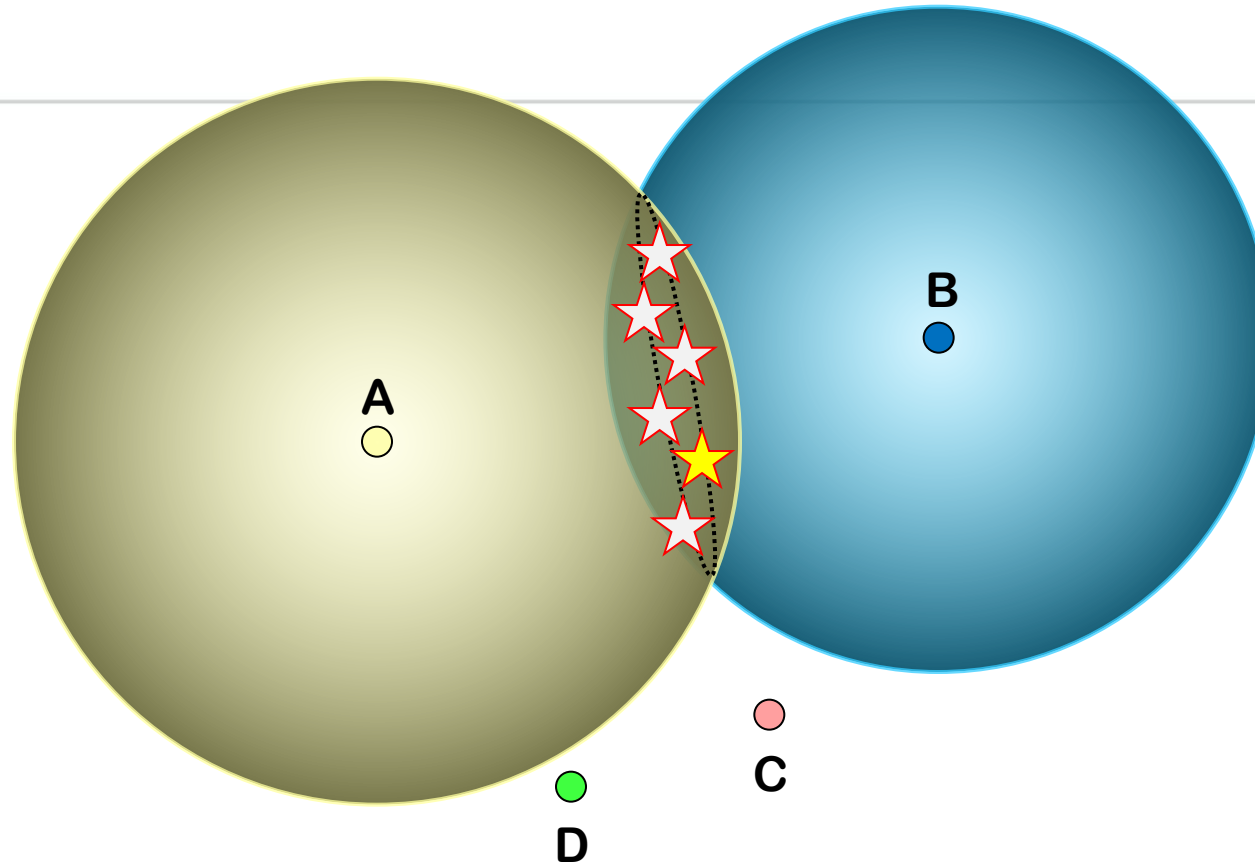
3D user position



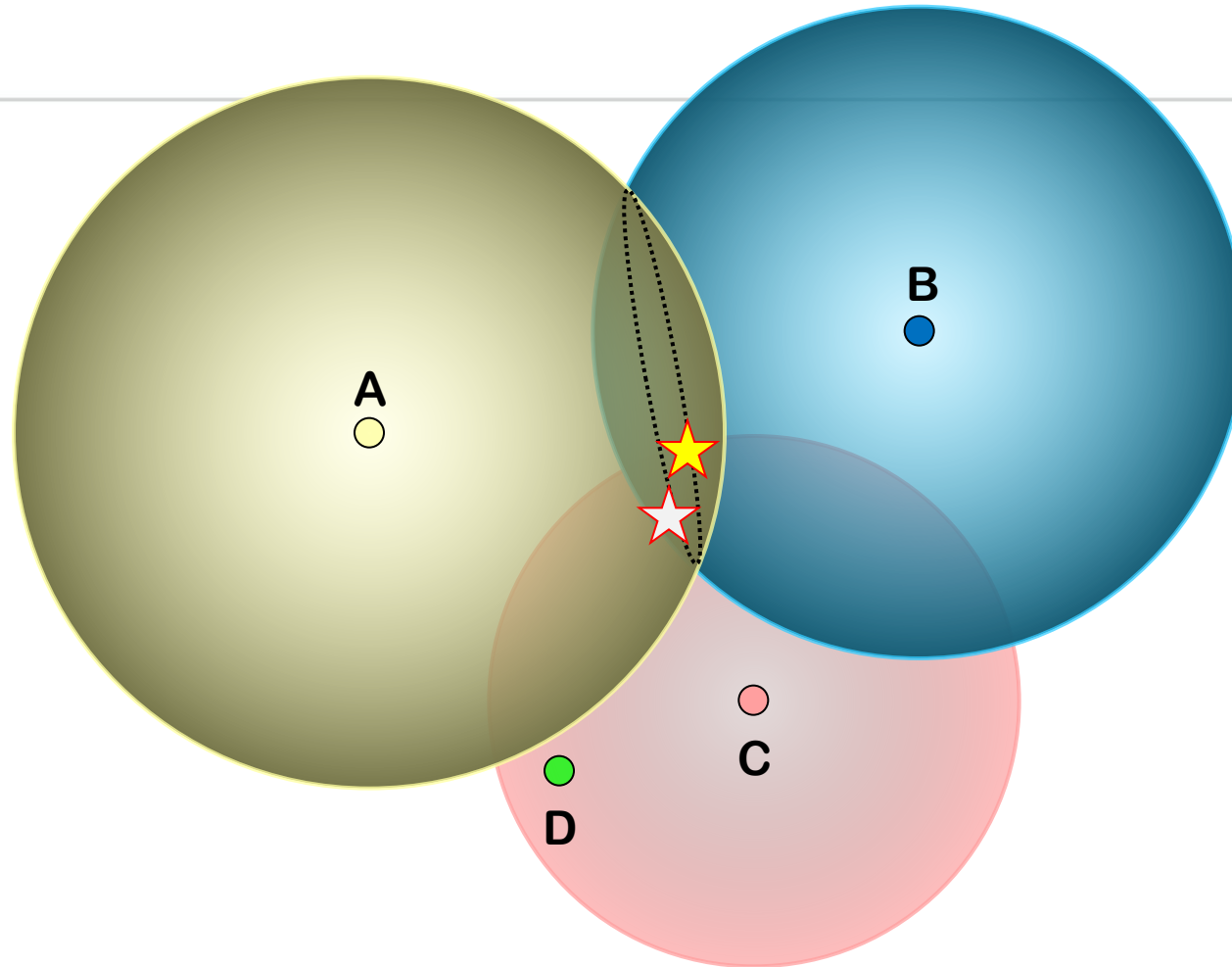
3D user position



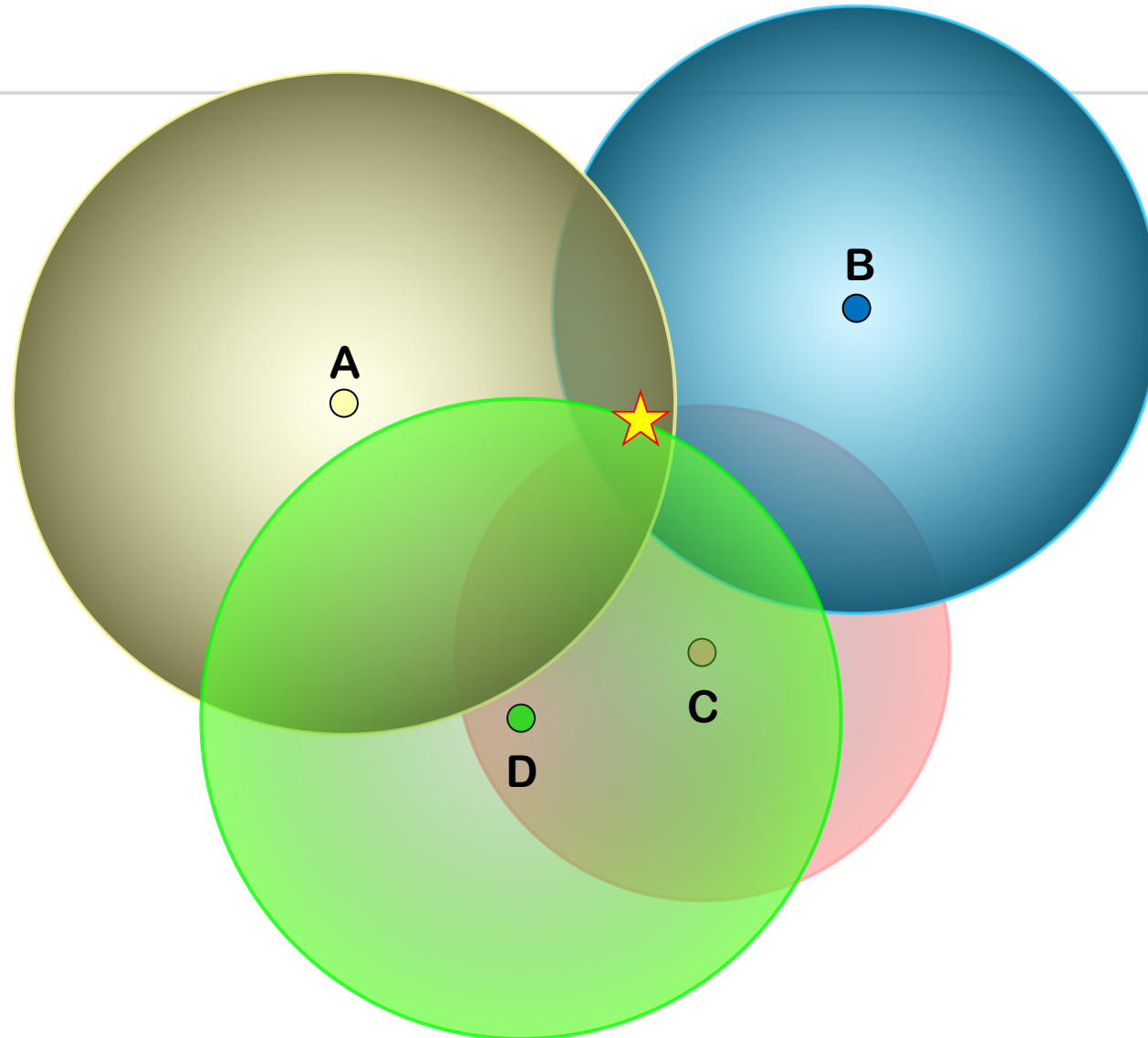
3D user position



3D user position

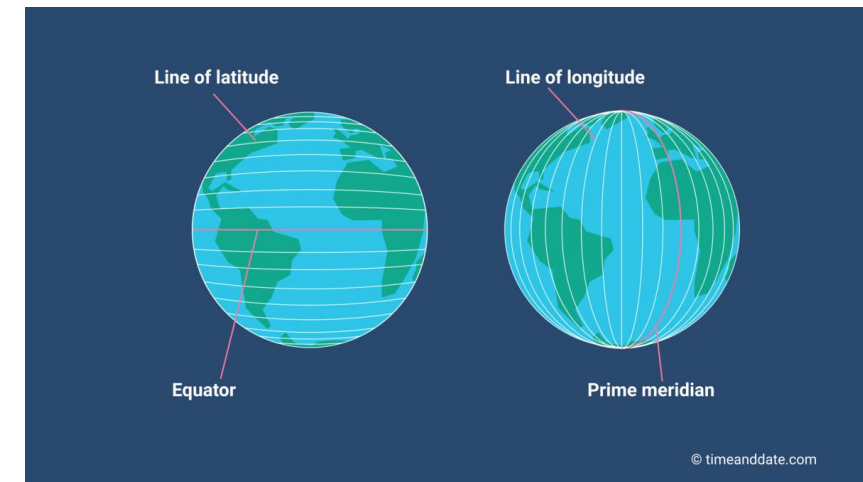


3D user position



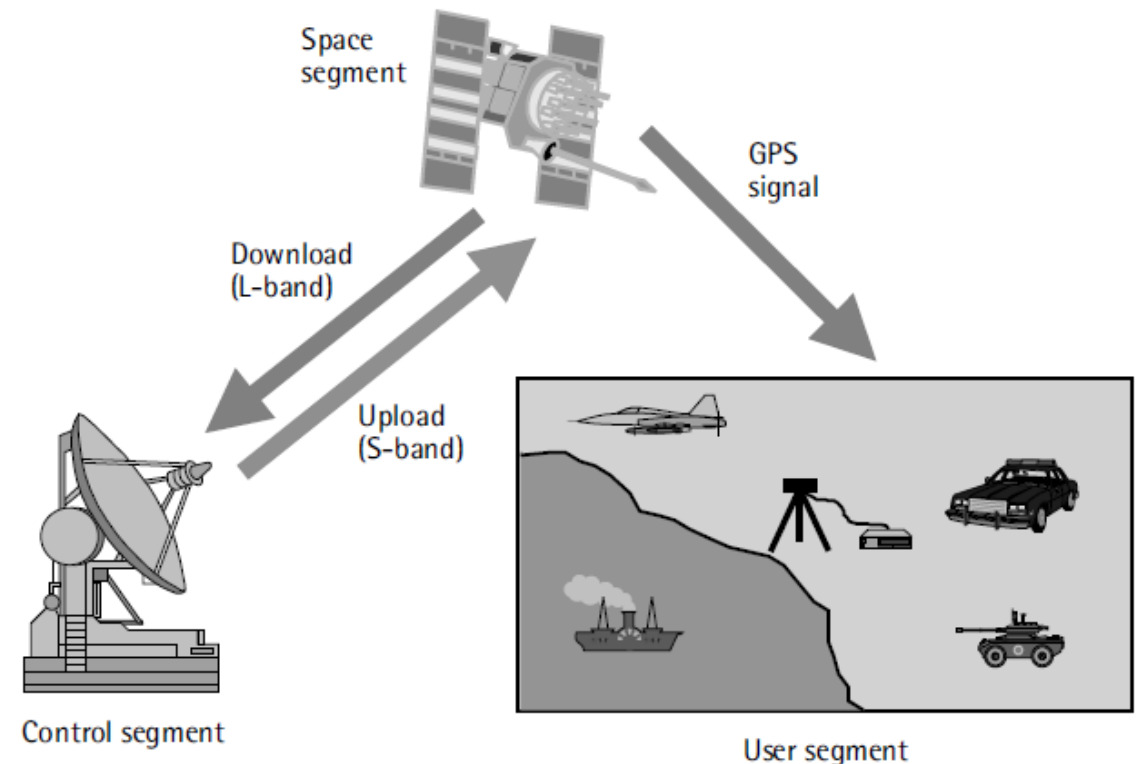
GPS - User position

- A GPS receiver must be locked on to the signal of **at least three satellites** to calculate a **2D position** (latitude (纬度) and longitude (经度) and track movement.
- With **four or more satellites** in view, the receiver can determine the user's 3D position (**latitude, longitude and altitude** (高度)).
- Once the user's position has been determined, the GPS unit can calculate other information, such as **speed, bearing, track, trip distance, distance to destination, sunrise and sunset time** and more.



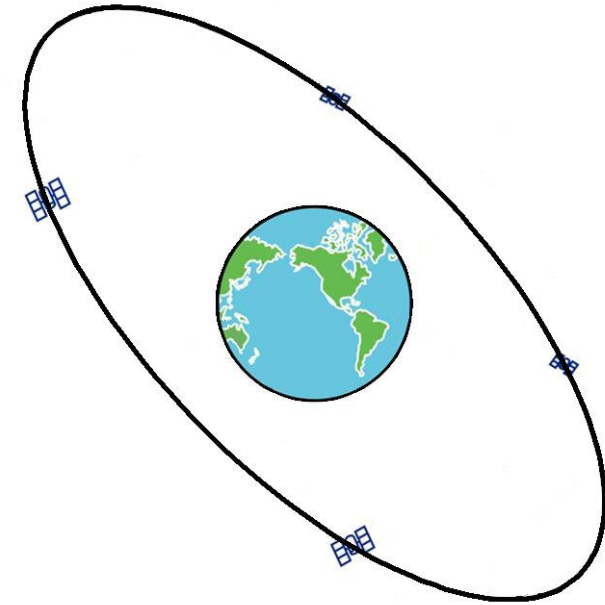
GPS Segment

- **Space segment:** the satellites that orbit the Earth, and the radio signals that they emit.
- **Control segment:** the ground stations that monitor and maintain the space segment components.
- **User segment:** the users with their hard and software to conduct positioning.



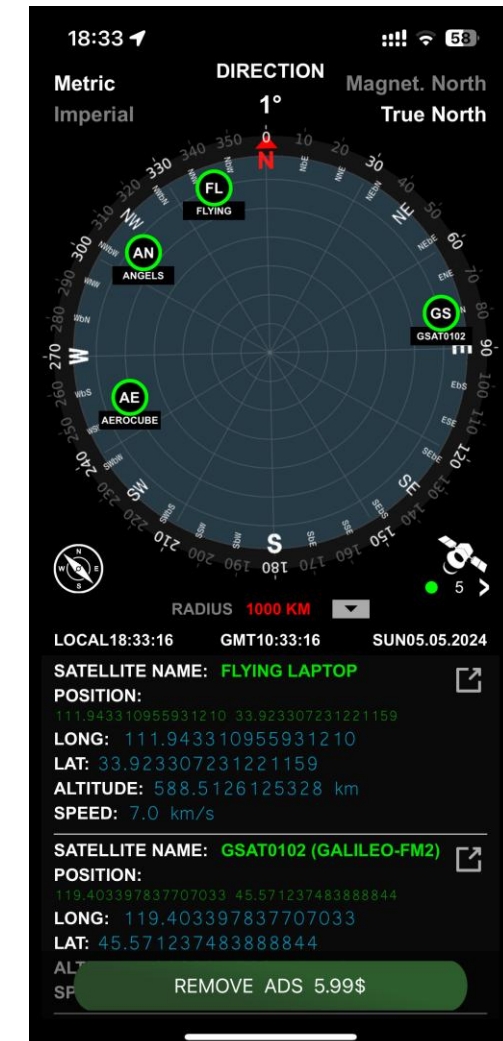
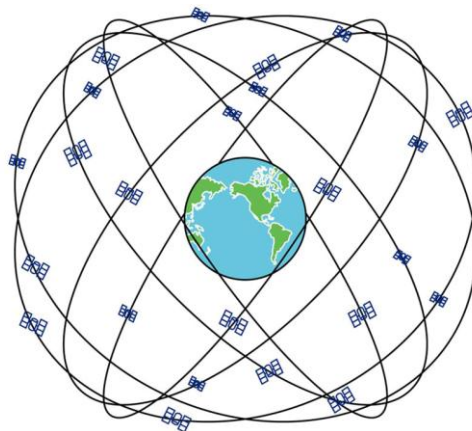
Space segment of GPS system

- The space segment of GPS consists of 24 satellites on 6 orbits (approx. 22,000 km from the center of the Earth):
- Each satellite carries a **clock**.
- Each satellite completes 2 orbits/day.
- Orbital period of **11 hr 58 min**
- 24 hour complete GPS coverage anywhere on the Earth.
- **Five to eight satellites are usually visible from any point on Earth**



The Space Segment

- Geometry ensures that **4 to 10** satellites are always visible from almost every place on Earth
- Arranged in **six orbital planes** with a **55-degree inclination** to ensure coverage over the entire planet



Orbital Configurations and Their Rationale - Why Certain Orbits Are Chosen

- Medium Earth Orbits
 - Most navigation satellites, including those for GPS and GLONASS, are in MEO at about 20,000 km altitude
- Rationale for MEO
 - Provides a good compromise between the geographic coverage of each satellite and the quality of the signal received on Earth
 - Enables a relatively small constellation to provide continuous global coverage with minimal signal delay that is acceptable for real-time navigation
- Orbit Period
 - Satellites in MEO typically complete an orbit in roughly 12 hours, meaning they appear in roughly the same position in the sky every 12 hours from any given point on Earth

Comparison of Different Satellite Orbits - Benefits and Drawbacks

- Low Earth Orbit
 - Benefits: Reduced signal delay and lower launch costs; ideal for earth observation satellites
 - Drawbacks: Requires a large number of satellites for continuous coverage due to the lower altitude and faster orbital speed
- Medium Earth Orbit
 - Benefits: Used by major GNSS providers ; balances coverage area and signal strength effectively
 - Drawbacks: Higher launch costs and more complex spatial configuration than LEO
- Geostationary Orbit
 - Benefits: Satellites remain stationary relative to a fixed position on Earth, simplifying ground antenna systems
 - Drawbacks: Limited to equatorial coverage; poor performance at high latitudes; not suitable for global navigation systems which require polar coverage

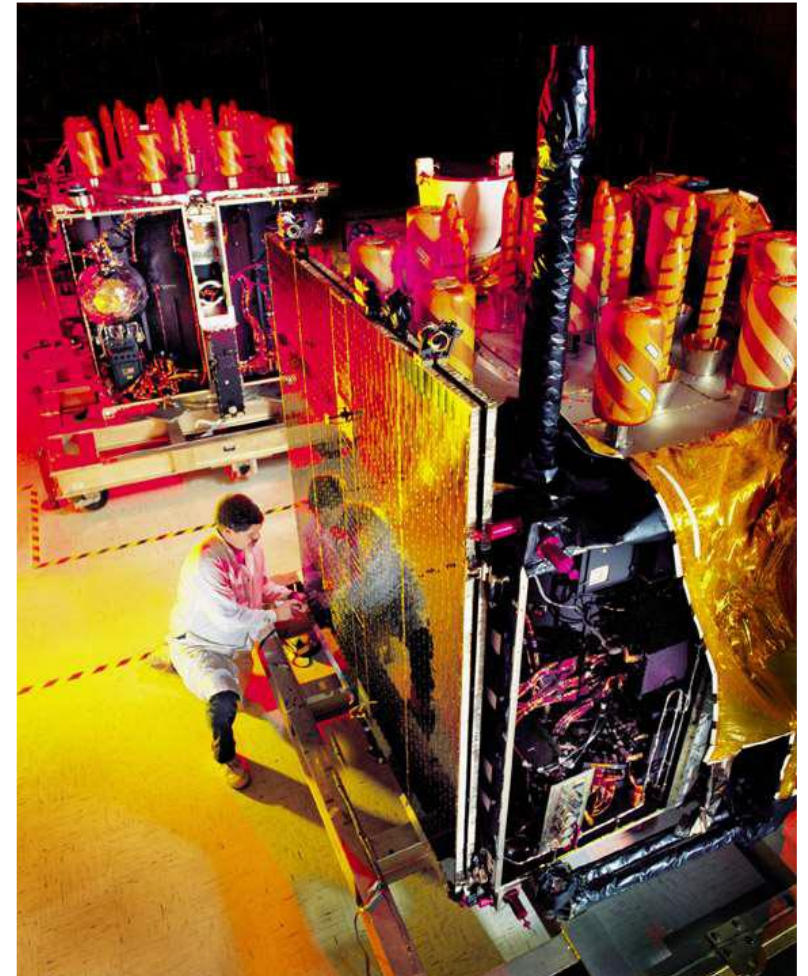
GPS Satellite Vehicle



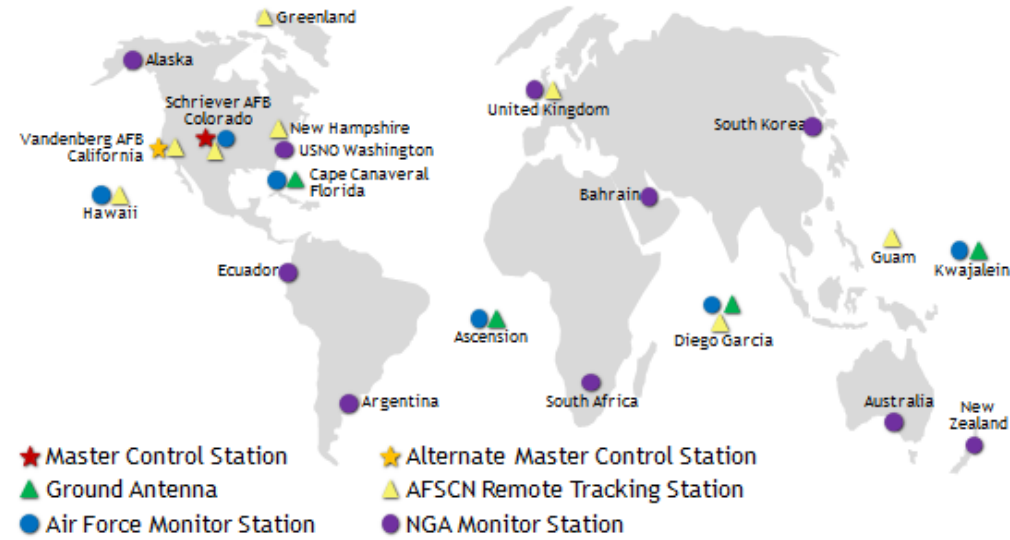
- Four atomic clocks
- Three nickel-cadmium batteries
- Two solar panels
 - Battery charging
 - Power generation
 - 1136 watts
- S band antenna—satellite control
- 12 element L band antenna—user communication

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- Weight
 - 2370 pounds
- Height
 - 16.25 feet
- Width
 - 38.025 feet including wing span
- Transmitted power is only 50 watts or less
- Design life—10 years
- Block IIR satellite



GPS - The Control & User Segment



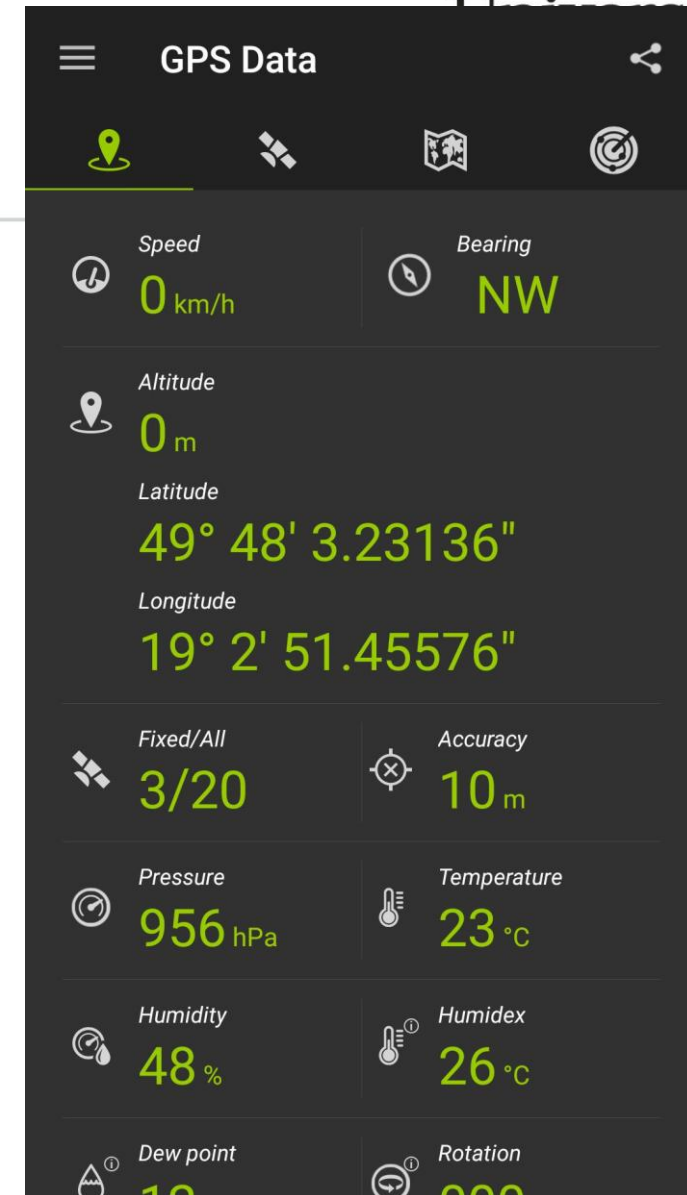
- Consists of a **global network of ground facilities** that **track** the satellites, **monitor** their transmissions, **perform analyses** and **send commands** and data back to the constellation.
- The current operational control segment includes 16 monitoring sites, a **master control station**, an **alternate master control station** and twelve ground antennas.

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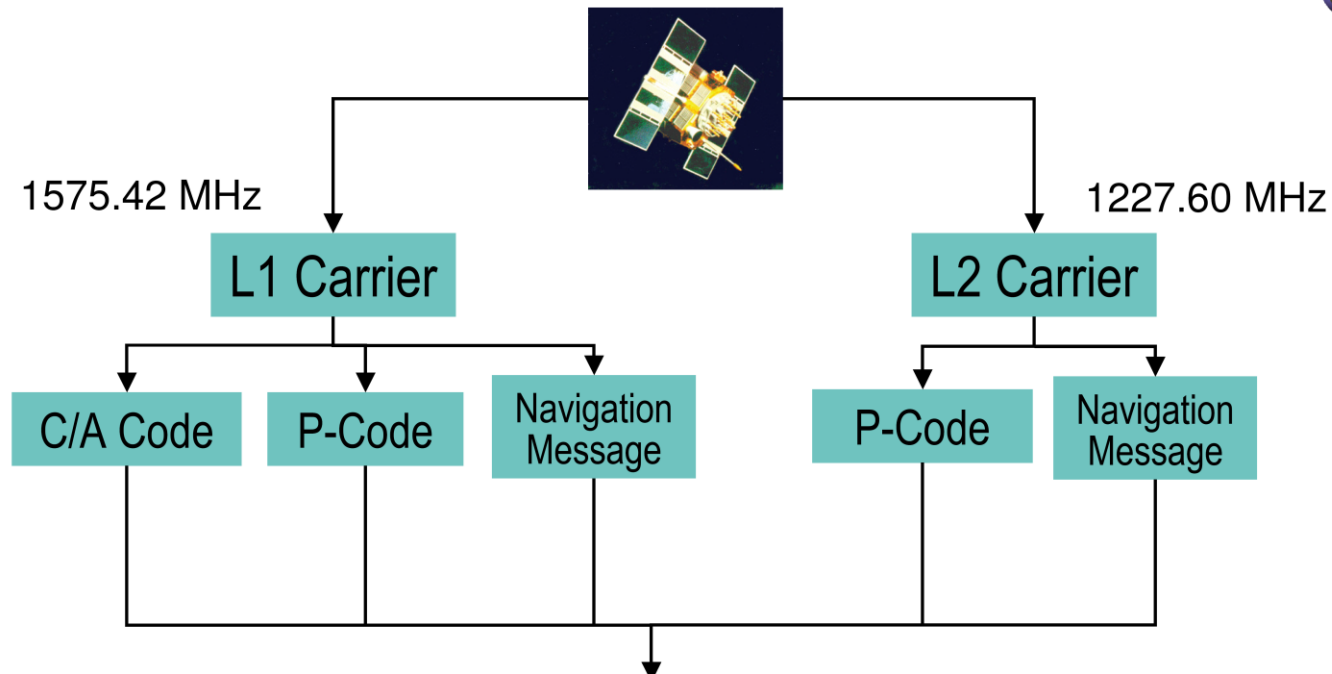
- **Monitoring sites** track the GPS satellites as they pass overhead and channel their observations back to the master control station. They also collect atmospheric data and navigation signals.
- The **master control station** determine and predict satellite location
 - System integrity
 - Behaviour of the satellite atomic clock
 - Atmospheric data and satellite Almanac
- **Ground antennas** are used to communicate with the GPS satellites for command and control purposes.

User segment

- GPS antennas & receiver/processors
- Position
- Velocity
- Precise timing
- Used by
 - Aircraft
 - Ground vehicles
 - Ships
 - Individuals



GPS Signal Structure



Each satellite transmits its own unique code

- Two frequencies used
 - L1 Carrier 1575.42 MHz
 - L2 Carrier 1227.60 MHz
- Codes
 - CA Code use L1 (civilian code)
 - P (Y) Code use L1 & L2 (military code)

C/A Code: A pseudorandom noise code that modulates the L1 carrier, making it easier for GPS receivers to lock onto

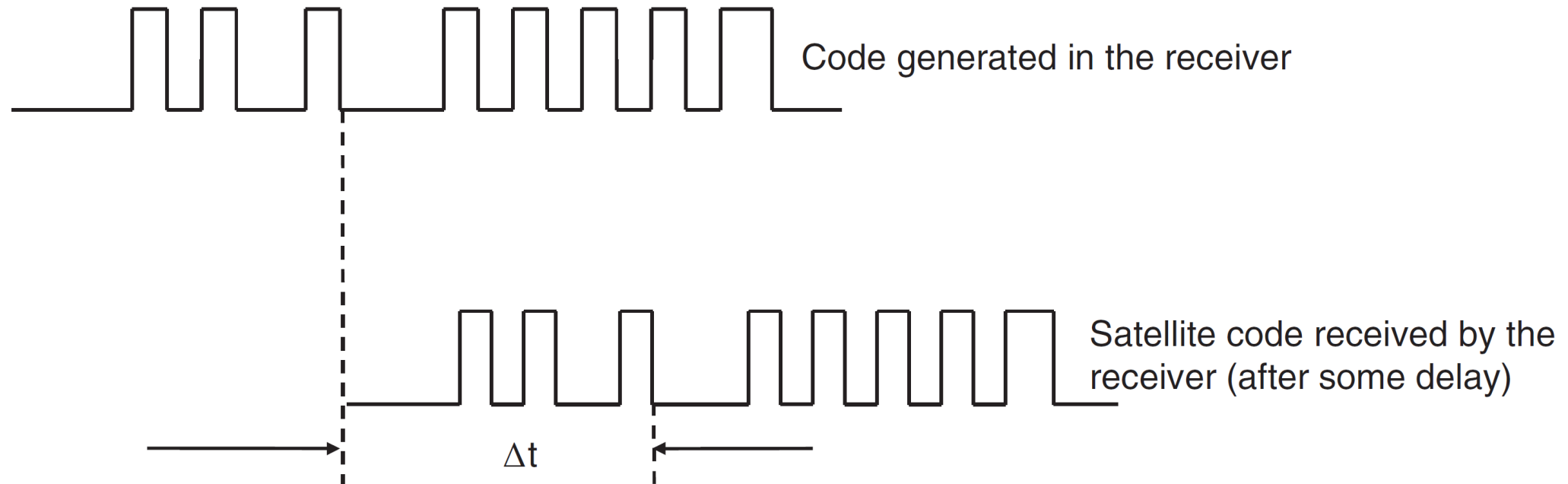
P Code: A more complex code that provides greater precision and resistance to jamming, used by the military

Navigation Message: A low bit-rate message that includes data on the satellite's time and status, its precise orbital parameters, and an overview of the state of the entire satellite constellation

Signal Structure of GLONASS - Differences from GPS

- Frequency Division Multiple Access
 - Unlike **GPS's CDMA**, **GLONASS** satellites transmit signals on different frequencies, a technique known as **FDMA**
- Frequencies Used
 - L1 Band: **Carries the FDMA signals using frequencies from 1598.0625 to 1605.375 MHz**
 - L2 Band: Uses frequencies from **1242.9375 to 1249.6875 MHz**, similar to GPS's L2, for enhanced services
- Code and Modulation
 - Uses FDMA combined with a similar **C/A code for civilian use and a P-code for military applications**
- Impact of FDMA
 - **FDMA allows for simpler receiver design** but can lead to frequency management challenges and **increased vulnerability to interference between frequencies**

Pseudorange Measurements



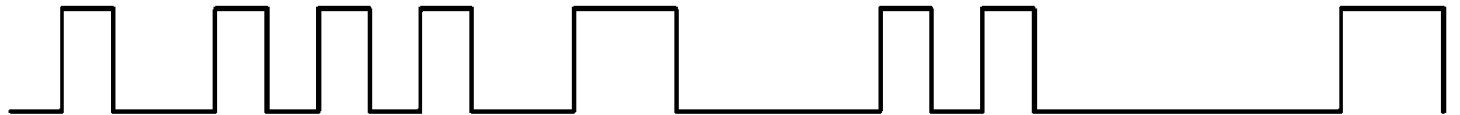
Δt is the time taken by the satellite code to reach the receiver
Pseudorange = $c \cdot \Delta t$

Pseudorange measurements

Example

- The code pattern generated by a transmitter is given in Figure below. **The same pattern is also generated in the receiver.** The pattern of the transmitter is received by the receiver after some time delay. The receiver pattern and delayed received pattern are shown in the figure. **Calculate the distance between the transmitter and the receiver, if the bit rate is 1 Mbps.**

Code received by the receiver



Code generated in the receiver

Solution

- Let us assume that the bit pattern is received by the receiver after a time delay of (Δt) . As can be seen from the figure on previous slide, the bit pattern received by the receiver is shifted wrt the pattern stored in the receiver by a time equal to the time period of 18 bits. Therefore, $\Delta t = 18 \times \text{one bit interval}$. As the bit rate is 1 Mbps, the bit period is $1/1 \times 10^6 = 1\mu S$.
- Therefore the time taken by the pattern to reach the receiver is $18 \mu S$ and the distance between the receiver and the transmitter = $3 \times 10^8 \times 18 \times 10^{-6} = 5400m$. This is just a simple illustration of how the **pseudorange** calculations are done. However, in actual practice, **cross-correlation techniques** are used to calculate the time delay (Δt) .

How accurate is unaided GPS?

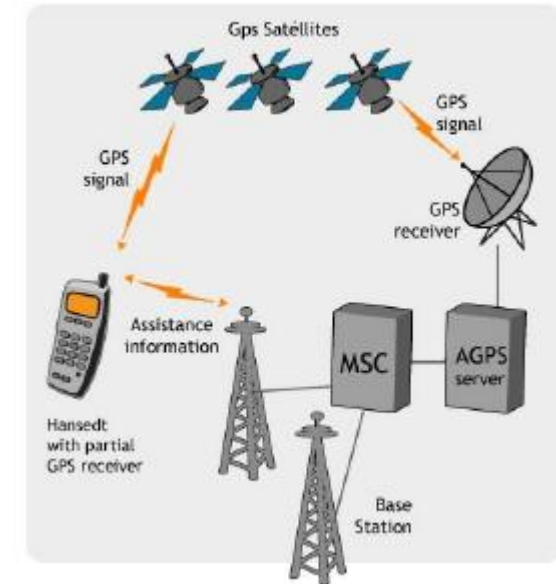
- Certain atmospheric factors and other sources of error can affect the accuracy of GPS receivers. **Most GPS receivers are accurate to within 15 meters on average.**

Garmin (major GPS receiver manufacturer) say...

“Today's GPS receivers are extremely accurate, thanks to their parallel multi-channel design. Certain atmospheric factors and other sources of error can affect the accuracy of GPS receivers. Garmin® GPS receivers are accurate to within **15 meters** (49 feet) 95% of the time. Generally, users **see accuracy within 5 to 10 meters** (16 to 33 feet) under **normal conditions.**”

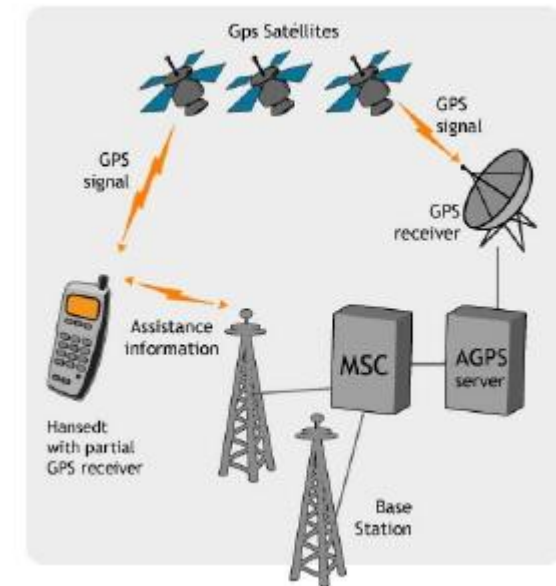
Assisted GPS on phones

- When we switch on Navigation app on our phones, it
 - First find orbit and clock data for the relevant satellites.
 - Can take anywhere from 30 seconds to 2 minutes.
- Cellular network tower have GPS receivers
 - Constantly pulling down satellite information and computing data



Assisted GPS on phones

- Assisted GPS on the phone pull this data from the tower and provides
 - Faster location acquisition
 - Less processing power is required by the device
 - Saves battery life
 - Location acquisition indoors

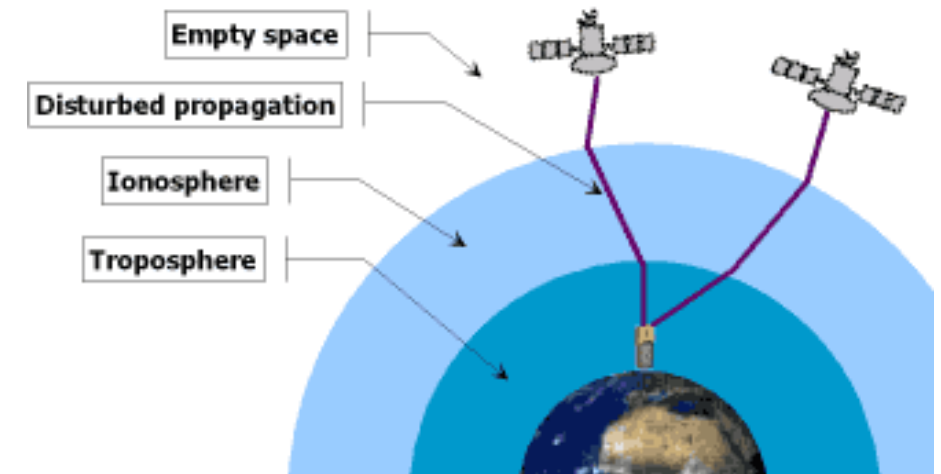


Sources of GPS signal errors

Factors that can degrade the GPS signal and thus affect accuracy include the following:

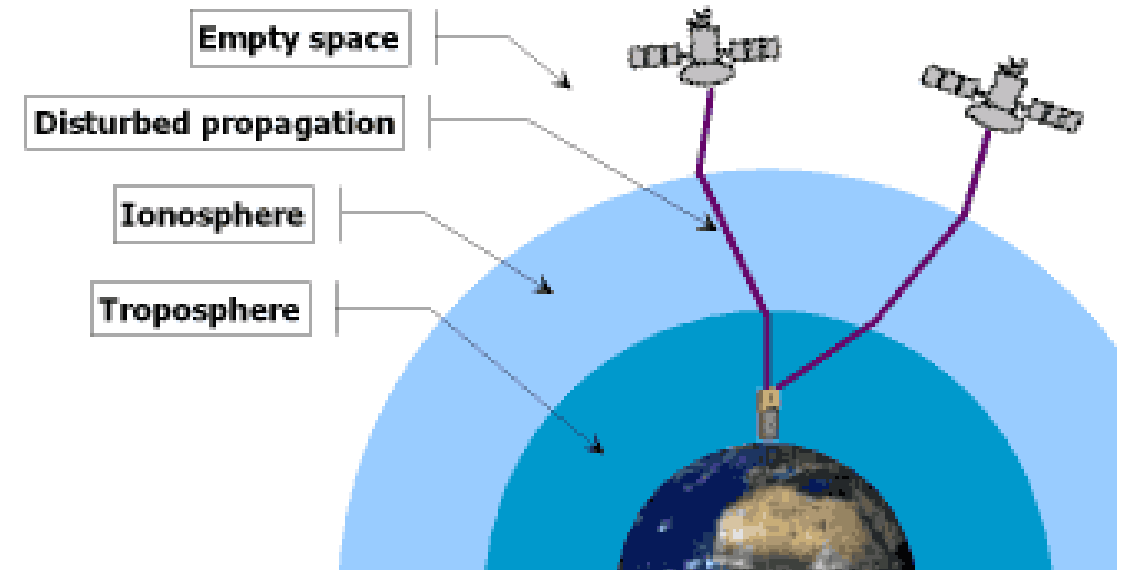
Propagation factors

- **Ionosphere and troposphere delays:** The satellite signal slows as it passes through the atmosphere. The GPS system uses a built-in model that calculates an average amount of delay to partially correct for this type of error. Typical error range : 1-10 m



Sources of GPS signal errors

- **Signal multi-path:** This occurs when the GPS signal is reflected off objects such as tall buildings or large rock surfaces before it reaches the receiver. This increases the travel time of the signal, thereby causing errors. Typical error : 0.5 m



Other factors influencing accuracy

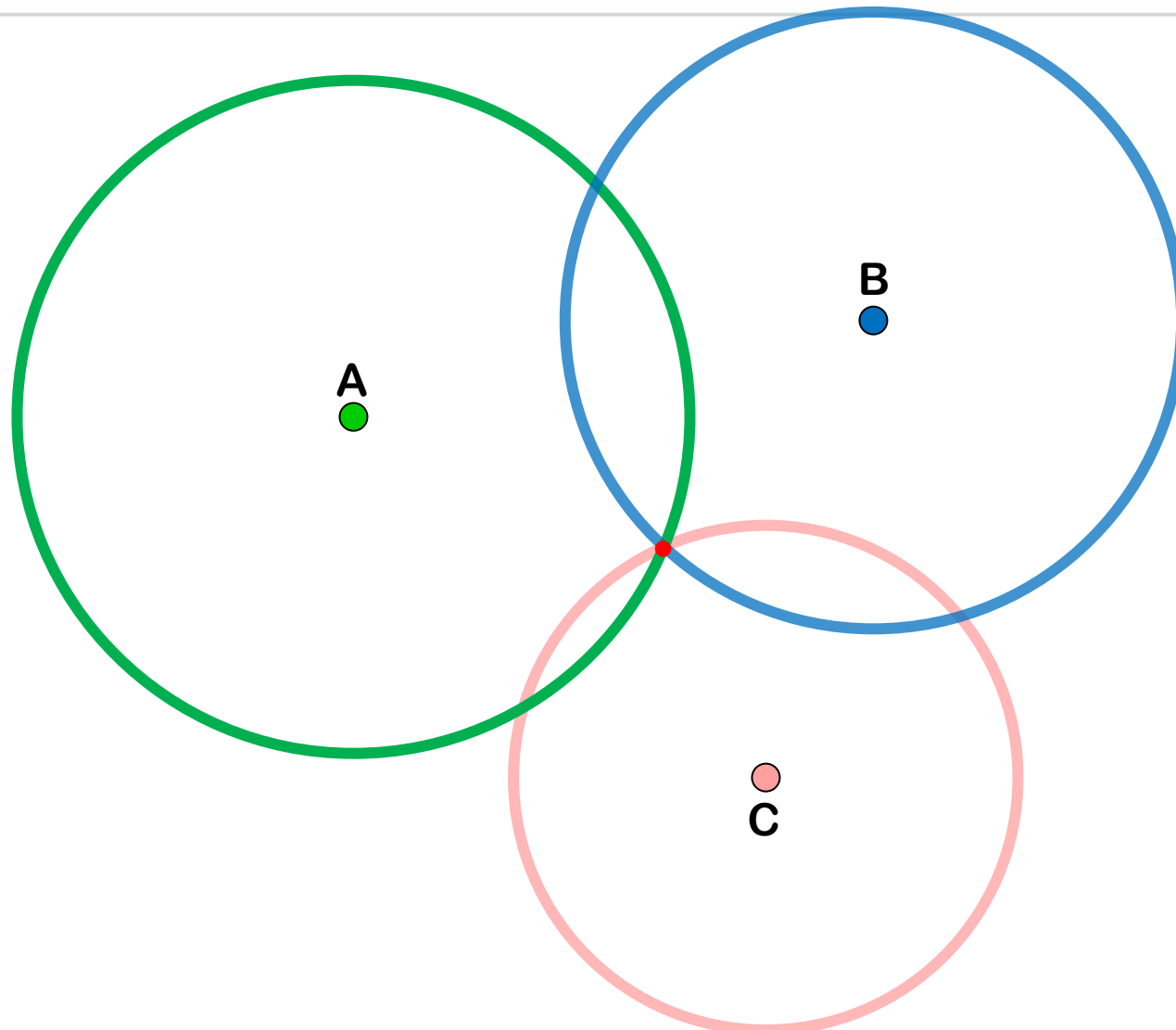
- **Orbital errors:** Also known as ephemeris (历书) errors, these are inaccuracies of the satellite's reported location. **Typical error: 1 m**

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- **Number of satellites visible:** The more satellites a GPS receiver can 'see', the better the accuracy. Buildings, terrain, electronic interference, or sometimes even dense foliage (叶子) can block signal reception, causing position errors or possibly no position reading at all. GPS units typically will not work indoors, underwater or underground.

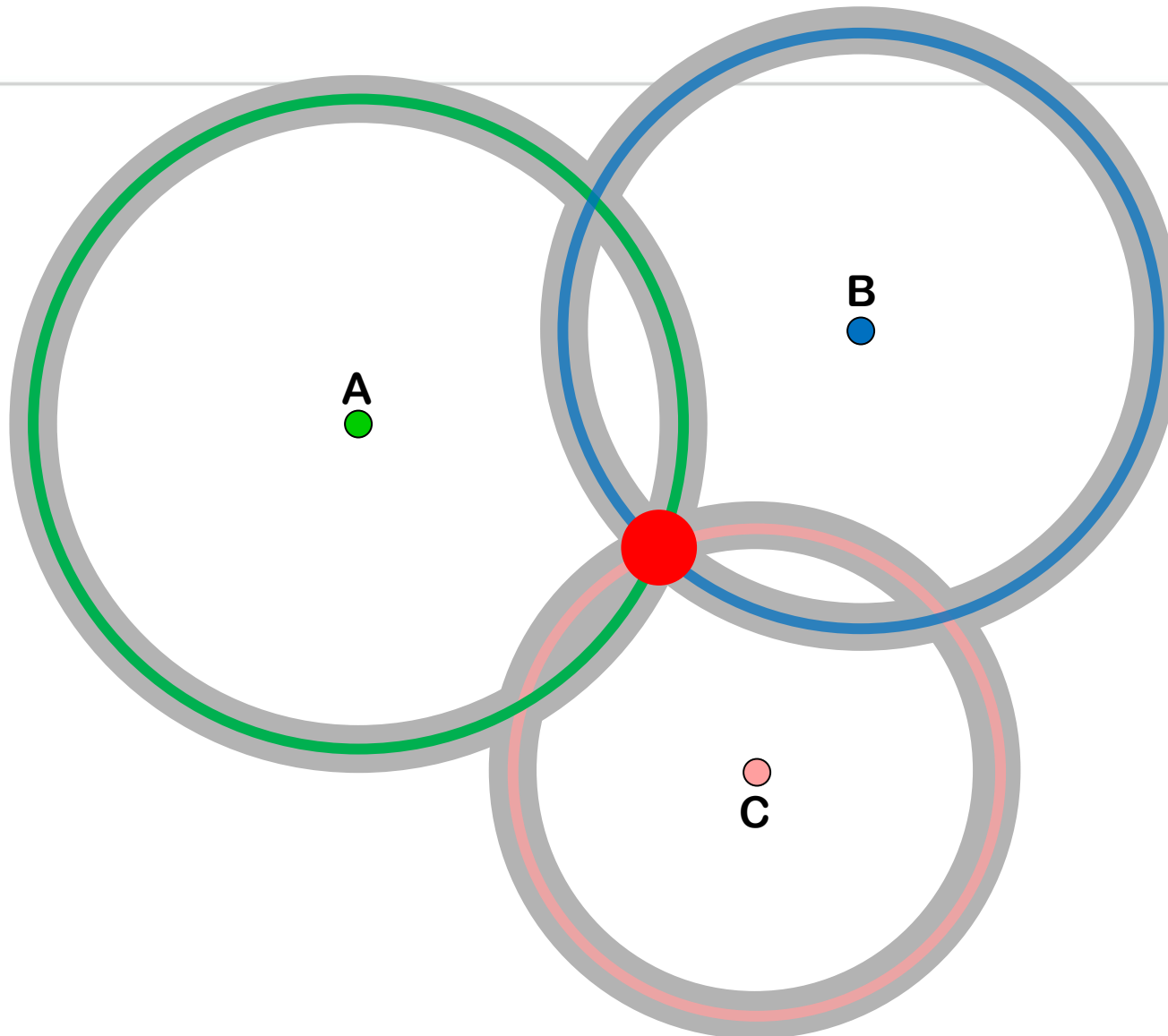
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- **Receiver clock errors:** A receiver's built-in clock is not as accurate as the atomic clocks onboard the GPS satellites. Therefore, it may have very slight timing errors. Typical error: 1 m

No receiver clock errors



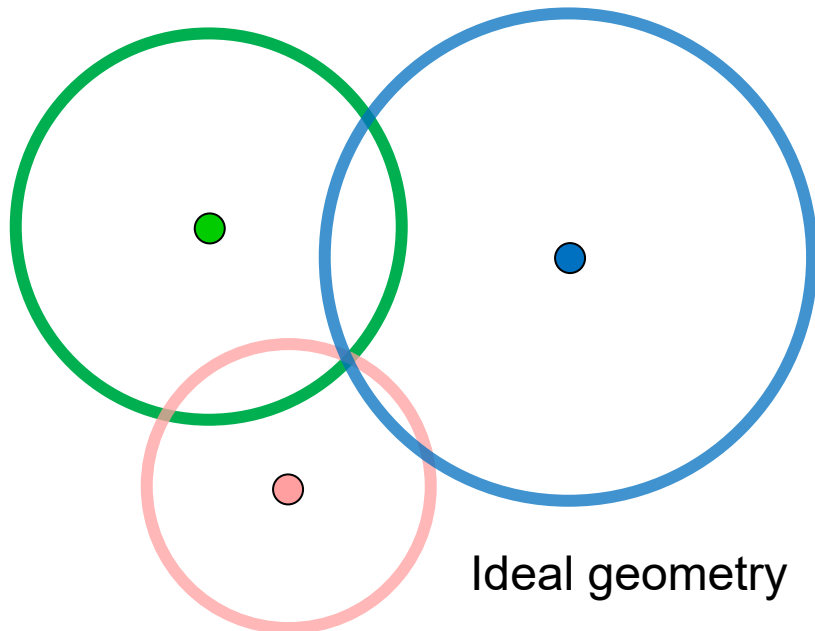
Receiver clock errors



Other factors influencing accuracy

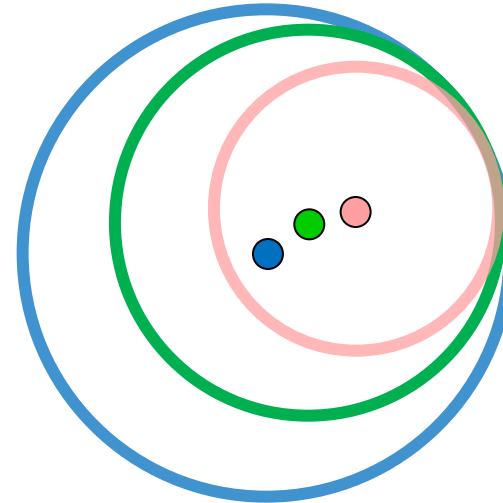
Satellite geometry/shading: This refers to the relative position of the satellites at any given time.

- **Ideal** satellite geometry exists when the satellites are located at **wide angles relative to each other**.



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- **Poor** geometry results when the satellites are located in a line or in a tight grouping.



Poor geometry

Cont'd

- **Intentional** degradation of the satellite signal
- “**Selective Availability**” (SA) is an intentional degradation of the signal once imposed by the U.S. Department of Defence.
- SA was turned off in May 2000, which significantly improved the accuracy of civilian GPS receivers.
- Could be turned back on?



Benefits of System Integration - Why Integrate?

- **Enhanced Coverage and Accuracy**
 - Integrating GPS and GLONASS can provide users with more satellite signals, improving positioning accuracy, especially in challenging environments like urban canyons or mountainous regions
- **Increased Reliability**
 - Redundancy offered by using two systems enhances reliability; if one system becomes temporarily unavailable, the other can still provide location data
- **Improved Global Performance**
 - Combining GPS with GLONASS ensures better performance at higher latitudes, where GPS alone might have less coverage
- **Cross-System Innovations**
 - Development spurred by integration leads to technological advancements that benefit all users, including more sophisticated multi-constellation receivers

Technical Challenges of Integration - Overcoming Differences

- **Frequency and Signal Discrepancies**
 - GPS and GLONASS use different frequencies and signal structures , requiring sophisticated receivers capable of handling both systems
- **Time System Alignment**
 - GPS and GLONASS operate on different time systems , necessitating complex calculations to synchronize these times in multi-constellation receivers
- **Orbital Dynamics**
 - The two systems have satellites in different orbital planes and altitudes, which can complicate the prediction models used by receivers to calculate positions
- **Interference Management**
 - Ensuring that signals from one system do not interfere with the other, especially in integrated devices, requires careful design and filtering technologies

Case Studies of Integration Success - Real-world Examples

- **Aviation Navigation**
 - Aircraft equipped with GPS/GLONASS dual-system receivers have reported **improved accuracy and reliability in navigation, especially during approaches and landings in difficult weather conditions**
- **Automotive Industry**
 - Car navigation systems that use both GPS and GLONASS can provide **more consistent and accurate positioning in urban settings**, improving user experience and safety
- **Scientific Research in Remote Areas**
 - Expeditions in the **Arctic and Antarctic regions benefit significantly from using both systems**, where GLONASS provides stronger signals due to better satellite visibility at high latitudes

Case Studies of Integration Success - Real-world Examples

- Emergency Response
 - Enhanced precision in crisis scenarios, aiding first responders in locating victims more quickly and accurately, potentially saving lives

Impact on Accuracy and Reliability - Quantitative Improvements

- **Statistical Improvements**
 - Studies have shown that using GPS and GLONASS in tandem can **improve positioning accuracy by up to 30-40%**, depending on the geographic location and environmental conditions
- **Temporal Reliability**
 - Integration **reduces the likelihood of signal loss**
- **Consistency Across Different Terrains**
 - Multi-constellation systems **perform better across varied terrains**, providing reliable services in urban, rural, forested, or mountainous areas without significant loss in accuracy
- **Future Projections**
 - Ongoing improvements in satellite technology and ground infrastructure are expected to **further enhance the benefits of integrating GPS and GLONASS**, promising even greater accuracy and reliability for all users

Next-Generation Satellite Systems: GPS III and Beyond

- **GPS III Enhancements**
 - GPS III satellites offer improved accuracy, enhanced security against jamming and spoofing, and increased overall system resilience
 - Equipped with a new L1C civilian signal compatible with other international systems like Galileo for improved interoperability
- **Extended Lifespan**
 - Designed for a 15-year operational lifespan, which is 25% longer than the previous blocks of GPS satellites
- **Future GPS III Follow-On**
 - Introduction of additional capabilities such as a regional military protection capability designed to increase anti-jam support in specific geographic areas
 - Implementation of a fully digital payload that offers flexibility in signal management and power allocation

Global Expansion of GLONASS: Plans and Prospects

- **Modernization Efforts**
 - Ongoing upgrades with the GLONASS-K2 and upcoming GLONASS-KM satellites that feature better accuracy and extended operational life
 - Introduction of new civil frequencies to improve service for non-military users
- **Increased Accuracy**
 - Targeting sub-meter level accuracy through improved satellite hardware and ground control enhancements
- **International Collaboration**
 - Expanding global partnerships to increase the adoption and integration of GLONASS services with regional augmentation systems

Emerging Global Systems: European Galileo and Chinese BeiDou

- **Galileo System**
 - Aimed at providing high-precision navigation services globally with better accuracy than existing systems
 - Offers unique features like the Open Service , Commercial Service , and Search and Rescue service
- **BeiDou System**
 - China's BeiDou navigation satellite system provide a global service since July 2020.
 - Offers dual-frequency navigation services to reduce dependence on foreign systems in critical infrastructure
- **Strategic Importance**
 - Both systems underscore the strategic importance of maintaining an independent satellite navigation capability for economic and security reasons

Interoperability of Systems: Making Them Work Together

- **Benefits of Multi-Constellation Receivers**
 - Combining signals from multiple systems like GPS, GLONASS, Galileo, and BeiDou increases accuracy, reliability, and availability worldwide
- **Challenges in Interoperability**
 - Aligning technical parameters such as signal structure, frequency bands, and synchronization of time and coordinate systems
- **Standardization Efforts**
 - International bodies and agreements are working towards global standards to ensure seamless interoperability among different satellite navigation systems

Recap- we talked about ...

- **The Global Positioning System (GPS)**
 - Space Segment
 - Control Segment
 - User Segment
- **Determining the position of a user (2D/3D)**
- **Assisted GPS**
- **Sources of signal errors**
 - Propagation factors
 - Orbital errors / Clock errors
 - Number of visible satellites and/or geometry
 - Intentional degradation (selective availability)